

Research Statement

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My areas of specialization are moral and political philosophy. My research in these areas includes my work on how requests create reasons; two research projects on the ethics of AI—one about structural injustice and another about causal models of discrimination—and a novel semantics for deontic modals and intention ascriptions. As this summary might suggest, I am a generalist at heart: my work in value theory intersects with philosophy of action, philosophy of language, philosophy of race, and philosophy of science. I also developed a related research program, which lies in logic and concerns the difficulty of probabilistic and causal reasoning.

1. Reasons by Request

You are not required to stop and aid every stranger you encounter. Seeing a lost tourist, you can permissibly decide not to help, thinking to yourself, “I’ll help the next time I see someone in their shoes.” But this thought rings hollow when the lost stranger *requests* your help. Your normative situation has changed. You must now cite a reason for not helping or else provide assistance. Requesters wield what is, on reflection, a remarkable power: to create reasons at will. The central puzzle of requests is to explain how this is possible. How can requesters give us reasons to do what they want, even when they stand in no relation of authority over us? And what would be lost if we saw requests as merely reminding us of reasons that exist already, rather than creating new ones?

In [How Requests Create Reasons](#) (*Philosophy & Phenomenological Research*, 2026), I argue that we have reasons to grant requests when and because doing so improves our compliance with our duties. To show that my view explains important forms of unfairness that result from requests, I draw on my work with Léa Bourguignon ([How to Count Sore Throats](#), *Analysis*, 2025), which explains when and why fairness permits ties among competing interests to be broken.

I next consider how the values served by the power to request constrain its permissible use. In [Why Not Ask?](#) (*R&R, Ethics*), I argue that some requests are wrongful *precisely because they create reasons*. Creating a reason often generates a double-bind for the requestee: if they act as requested, they pay the practical cost of doing so; if they do not, they pay a moral cost, in the form of a reason for regret. While this double-bind does not always make requests wrongful, I show that in more serious cases it can be wrongfully intrusive, manipulative, and coercive.

I turn, finally, to a foundational question: How can the ability to create a reason by requesting be explained *by its value*? After all, it would be wishful thinking for us to infer that we possess the ability to fly, simply because it would be valuable if we had it. In **What Justifies Normative Powers?**, I argue that the power to create reasons by requesting can be justified by comparing its value *for each person*: a person possesses this power when, in general, it is better for *requesters* in these circumstances to *possess* the power than it would be for *requestees* to *not be subject* to the power. After generalizing my account to commands and promises, I show that it rests on a familiar and widely-accepted form of explanation in ethics, which justifies normative principles by considering what the world would be like for each of us, were those principles generally accepted.

2. Artificial Intelligence and Structural Injustice

My research on the ethics of AI, which spans explainable AI, algorithmic fairness, and AI alignment, concerns *structural injustice*: I am interested in understanding how several decisions, which appear innocent in isolation, can create a system that is unjust.

- My work in explainable AI is concerned with how, even if (say) race and gender do not influence a decision when taken in isolation, an algorithm can still discriminate on the basis of race or gender, taken together with other factors. With Harry Sha and Li-Yang Tan, I developed a method for producing explanations of automated decision making ([A Generalization of the Satisfiability Coding Lemma and Its Applications](#), *International Conference on Theory and Applications of Satisfiability Testing*, 2022), which resolves a problem posed by Quine over 70 years ago, and won the conference’s Best Theory Paper Award.
- In a paper published in the premier algorithmic fairness venue ([Multiplicative Metric Fairness Under Composition](#), *Foundations of Responsible Computing*, 2023), I show how different definitions of “similar treatment for similar individuals” can differentially determine whether *groups* of individuals are treated unfairly, and whether several decision-makers that are fair in isolation from each other can *compose into a system* which is unfair. I prove that the most popular definition of equality of opportunity for individuals fails to protect against structural injustice, whereas a definition that has been largely overlooked does so.
- Leading methods for alignment like Reinforcement Learning from Human Feedback train AI to align with individuals’ preferences, but doing so implicitly aggregates preferences in ways that may violate basic principles of fairness in social choice. With several co-authors, I proposed a framework for using social choice to align machine learning models’ choices with human values ([Social Choice for AI Alignment](#), *International Conference on Machine Learning*, 2024), which was the third most-cited paper in social choice in 2024. My interest in social choice extends beyond AI: in a paper with Wesley Holliday, Chase Norman, Eric Pacuit, and Cynthia Wang ([Stable Voting and the Splitting of Cycles](#), *Association for the Advancement of Artificial Intelligence*, 2026), we resolve a conjecture about the relationship between two major voting rules.

I also published two papers on artificial intelligence in leading computer science venues. [Conditional Negative Sampling for Contrastive Learning of Visual Representations](#) (*International Conference on Machine Learning*, 2021) explains why certain transformations on datasets (e.g. adding random noise) improve learning, while others (e.g. rotating an image) do not. [Zero-Shot Learning for Code Education](#) (*Association for the Advancement of Artificial Intelligence*, 2019) provides a method for “zero-shot learning,” i.e. learning without access to any examples. We use a fixed probabilistic grammar to generate samples, and the learner uses this simulated dataset instead of a real one. This paper won the conference’s Outstanding Student Paper Award (presented to 1 out of 7,095 submitted papers).

3. Causal Models for Social Constructions

Direct gender discrimination occurs when a person’s gender *causes* their worse treatment. This causal notion of discrimination is invoked by algorithmic fairness criteria, U.S. law, and correspondence studies in the social sciences, but it has proven difficult to spell out in a precise and plausible way. What does it mean to only vary gender and hold other attributes fixed, given that gender are bound up with so many features of a person? For example, suppose Maya was a Girl Scout, went to Barnard, became a nun, and is wearing a dress. What does it mean to only vary gender in this case? Should we imagine a former Boy Scout who went to Columbia, became a priest, and is wearing robes? A central problem in conceptualizing discrimination is to explain how it makes sense to think of protected attributes like gender and race as causes, given that these attributes are embedded in a nexus of social factors that resist isolated treatment.

Faced with this problem, some give up on modeling protected attributes like gender as social kinds, rather than strictly biological ones. Others suggest that we give up on traditional causal models for discrimination, which require the ability to isolate a cause like gender from other attributes. With Frederick Eberhardt, Thomas Icard, and Kara Schechtman, I developed a framework for modeling discrimination that gives us the best of both worlds: we explicitly model attributes like gender as social kinds, while preserving a precise formal model of discrimination. The key is to model protected attributes like gender as *high-level abstractions* of their many constituents: by explicitly representing both causal and constitutive relationships, we can model claims about discrimination as generalizations about the effects of many different *ways* of varying gender. In [Modeling Discrimination with Causal Abstraction](#) (under review), we introduce this framework, which is now being employed by social scientists to test for discrimination over real-world datasets. In an ongoing sequel co-authored with Kara Schechtman and Lydia Liu, we use our framework to design a novel test for discrimination in Large Language Models.

4. How Hard Are Probabilistic and Causal Reasoning?

Causal reasoning seems indisputably harder than purely “correlational” or probabilistic reasoning: it requires that we perform deliberate experimentation and that we accept strong assumptions needed to justify causal conclusions. Even so, in a precise sense, causal and probabilistic reasoning are equally difficult. Duligur Ibeling, Thomas Icard, and I propose that many of the most prominent (i.e. frequentist and Bayesian) approaches to statistical inference can be understood, at least in part, as attempts to turn an *inductive* problem into a *deductive* one. We show how to computationally reduce problems about causation to equivalent ones about correlation: the computational complexity of deductive probabilistic and causal reasoning is the same ([Is Causal Reasoning Harder than Probabilistic Reasoning?](#), *Review of Symbolic Logic*, 2022).

Several papers responding to our work extend our formal systems to include a special summation operator, a feature of Pearl’s do-calculus. We show that suitably restricted, the operator does not change our results, but that when the operator is interpreted in the standard way, the do-calculus becomes undecidable: there is no procedure for causal reasoning, as it is traditionally modeled ([On probabilistic and causal reasoning with summation operators](#), *Journal of Logic and Computation*, 2024). Positively, our work circum-

scribes the complexity of popular causal inference tools and shows that causal reasoning adds no computational cost. Negatively, we show that the standard way of formalizing the do-calculus makes probabilistic and causal reasoning impossible.

Logicians and psychologists often distinguish *qualitative* and *quantitative* probabilistic reasoning. But how exactly should this distinction be drawn? Duligur Ibeling, Thomas Icard, Krzysztof Mierzewski, and I use computational complexity to draw a robust distinction between qualitative and quantitative reasoning, which tracks independently important features of the corresponding logical systems, including finite axiomatizability ([Probing the Qualitative-Quantitative Divide](#), *Annals of Pure and Applied Logic*, 2023). In **Reasoning about Confirmation** (under review), which I presented at Carnegie Mellon’s 2025 Probability Logic Workshop, I strengthen our prior results, and along the way resolve some open questions about the difficulty of reasoning about when a piece of evidence confirms a hypothesis.

5. How Questions Structure Plans

Intention ascriptions often feature sequences of telic infinitivals:

- (1) John plans to go to the store to get thread to fix his jacket.

This ascription *distributes*, entailing:

- (2) John plans to go to the store.
- (3) John plans to go to the store to get thread.

But the *order matters*. (1) does not entail:

- (4) John plans to get thread to go to the tailor.

Further, intentions are *partly closed under implication*, since (3) entails (5) but not (6):

- (5) John plans to go somewhere to fix his jacket.
- (6) John plans that his jacket be damaged, such that it needs to be fixed.

In **How Questions Structure Plans**, Seth Yalcin and I propose a model of intention which predicts these data. At the heart of our model lies the idea that intentions pose and answer how-questions, such that (7) sounds fine, and (8) sounds uncooperative:

- (7) John has a plan for how to get thread.
- (8) John plans to go to the store to get thread. ??But how does he plan to get thread?

We model an agent’s planning state as a set of *teleologies*, where a teleology is a sequence of nested how-questions, for example:

$$\begin{aligned}\tau = & (\text{How to fix John's jacket?}, \\ & \text{How to get thread to fix John's jacket?}, \\ & \text{How to go to the store to get thread to fix John's jacket?})\end{aligned}$$

Thus (1) is true because τ belongs to John’s intention state, and τ *settles* that John will

go the tailor to get thread to fix his jacket. We show that our model recovers Beddor & Goldstein's recent semantics for simple ascriptions of the form " x intends to F " as a special case. Further, our semantics follows from a compositional semantics for purpose clauses "to F ", "to F to G ", etc., which gives us novel semantics for a range of teleological explanations, for example:

- (9) John went to the store to get thread to fix his jacket.
- (10) John ought to go to the store to get thread to fix his jacket.
- (11) The dishwasher ran its motor to splash suds to clean the dishes.
- (12) The seedling needs water to grow to reach the sunlight.

In sum, we introduce a wide swath of data about planning ascriptions and a model of intention that predictions them. This model follows from a compositional semantics for purpose clauses, which leads to a general semantics for teleological explanations.